Uday Dhawan

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.35
SRCD Saraswati Vidya Mandir, Biswan	Class XII (U.P. State Board)	2014	88.40%
SRCD Saraswati Vidya Mandir, Biswan	Class X (U.P. State Board)	2012	84.83%

SCHOLASTIC ACHIEVEMENTS

Excellence in Academics Award - IIT Hyderabad

PROJECTS

Microstructure-Based CPFEM of Deformation in LPBF Ti-6Al-4V: Integrated SEM/EBSD Validation | *MTech Thesis* | *ABAQUS* | *MATLAB* | *TSL-OIM*

- Characterized crystallographic texture & microstructure via **SEM/EBSD** at different points in tensile test.
- Built an EBSD-derived **RVE** using **MTEX (MATLAB)** & **DREAM.3D**, implemented the crystal plasticity model via a custom **UMAT subroutine** in **Abaqus**.
- Validated the CPFE model against experimental stress-strain response to predict **crack initiation sites** & **grain-to-grain plasticity.**

FE Thermal Modelling of LPBF Process | *ABAQUS* | *Fortron*

- Coded an Abaqus DFLUX Fortran subroutine for a moving Gaussian laser (150 W, = 0.36, 100 µm beam radius, 200 mm/**s** scan speed).
- Built a 3D transient heat-transfer model of a 2×2×0.2 mm SS316L plate with temperature-dependent properties, convection/radiation losses and a **10 µm graded mesh.**
- Predicted a **4540 K peak temperature** and a **~20 µm deep × 100 µm wide** melt pool for a single LPBF scan track.

Automated Ball Tracking and Velocity Estimation using MATLAB Circular Hough Transform | *MATLAB Image processing*

- Developed a MATLAB script using **Circular Hough Transform** (50-100 px radius, 0.92 sensitivity) to auto-track a ball across video frames.
- Calibrated pixels to metres with a 0.6 m reference and computed velocity by forward differencing, with the ball detected in **58 frames.**
- Analysed velocity-time data (peak 1.36 m/s) and **traced deviations** from theory to low-resolution video and detection dropouts.

Predicting EV Chassis Material Suitability using ML | *Python*

- Built an ML pipeline (Random Forest, ANN) to predict EV chassis material suitability from 1,452 samples, engineering 7 features via correlation analysis and RFE
- Tuned models with GridSearchCV (**5-fold CV**), achieving **R^2 = 0.978** and **RMSE = 0.220** on validation data with the ANN
- Validated predictions on 100 unseen materials, reaching **90%** agreement between RF and ANN outputs

ML Pipeline for Predicting Mechanical Properties of Iron Alloys | *Python*

- Built regression pipeline in Python (pandas, scikit-learn) to predict UTS and Elastic Modulus for ~1200 iron alloys from chemical composition and property data.
- Engineered features (mean/range from min-max pairs, alloy-type encoding, correlation filtering at r>0.95) and handled missing data via median imputation across 50+ raw columns.
- Benchmarked Linear, Ridge, Random Forest, and Gradient Boosting Regressors using MultiOutputRegressor and 5-fold cross-validation, achieving avg. R^2 of 0.96 with Gradient Boosting.

Development of Bicycle-Operated Water Purification System for Sustainable Clean Water Generation | *BTech Final Year Project* | *SOLIDWORKS*

- Designed & fabricated a pedal-powered, off-grid water purifier (dynamo + battery + diaphragm pump) - total cost **8,590.**
- Engineered a 12V diaphragm pump system delivering **5.5 L/min** through 2-stage sediment & carbon filtration.
- Validated output water via IS 3025 lab testing: **pH 7.31**, hardness **150 mg/L** - well within potable limit (<200 mg/L).

RELEVANT COURSES

Advanced FEM, Topology Optimization, Design for Metal Additive Manufacturing, Machine Learning & its application

INTERNSHIP & COURSES

Def-Space Summer Internship | BSERC (Remote) | Convolution Neural Network

- Gained interdisciplinary exposure to AI/ML application in the field of Defence, Advanced Engineering.
- Worked on a CNN project in which built a baseline CNN with **6** convolutional layers and **3** max-pooling layers trained on CIFAR-10 (**10 classes, 60,000 images**).
- Achieved **84.94%** accuracy and F1-score of **0.83**, outperforming a VGG16 transfer learning model which gave only **56.82%** accuracy.

Case Knights - Consulting & Product Management Series | Finance & Consulting Club, IITH

- Gained hands-on exposure to Root Cause Analysis, Guesstimates, and Product Management through case-based learning.

SKILLS

CAE Tools: ABAQUS, ANSYS, HyperWorks Optistruct, SOLIDWORKS, AutoCAD

Material Characterization Tools: TSL OIM, MTEX & STABIX (MATLAB), DREAM.3D

Programming: MATLAB, Fortran (User Subroutine), Python

Libraries & Tools: OpenCV, LaTeX, Overleaf

Data Analysis & Visualization: Microsoft Office

CERTIFICATIONS

MathWorks Training Services: MATLAB Onramp, ML Onramp, Intro to FEA with MATLAB, Explore CNNs

AutoCAD-ME (40 hrs), Xtrude Engineers' CAD

POSITION OF RESPONSIBILITY

Volunteered in Office of Carrier Services at IIT Hyderabad

- Coordinated and managed campus recruitment events, ensuring seamless execution of schedules and activities.
- Worked collaboratively with a team to facilitate placement activities, demonstrating strong cooperation and teamwork.

EXTRACURRICULAR

Achieved **Gold Medal** in Hockey - DIESTA, IIT Hyderabad

Achieved **1st Place** in Nautankishala Event - GK (DU), Haridwar